

A Compact Memristor-based Dynamic Synapse for Spiking Neural Networks

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Abstract—Recent advances in memristor technology lead to the feasibility of large-scale neuromorphic systems by leveraging the similarity between memristor devices and synapses. For instance, memristor cross-point arrays can realize dense synapse network among hundreds of neuron circuits, which is not affordable for traditional implementations. However, little progress was made in synapse designs that support both static and dynamic synaptic properties. In addition, many neuron circuits require signals in specific pulse shape, limiting the scale of system implementation. Last but not least, a bottom-up study starting from realistic memristor devices is still missing in the current research of memristor-based neuromorphic systems. Here, we propose a Memristor-based Dynamic (MD) synapse design with experiment-calibrated memristor models. The structure obtains both the static and dynamic synaptic properties by using one memristor for weight storage and the other as a selector. We overcame the device nonlinearities and demonstrated spike-timing-based recall, weight tunability, and spike-timing-based learning functions on MD synapse. Furthermore, a temporal pattern learning application was investigated to evaluate the use of MD synapses in spiking neural networks, under both spike-timing-dependent plasticity (STDP) and Remote Supervised Method (ReSuMe) learning rules.

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I. INTRODUCTION

Spiking neural networks (SNNs) increases the level of realism in neural simulations by incorporating time into neuron/synapse models [1]. Since information is encoded by the precise timing of spikes, a train of spikes can contain more information than binary/rate encoding methods. Moreover, SNNs minimize the data communication cost by distributing data into local memories (*i.e.*, synapses) close to the associated computing units (*i.e.*, neurons) throughout entire system. SNNs can reduce the communication cost and consequently may achieve brain-level computing efficiency [2]. For example, *TrueNorth* – the latest spike-timing-based neuromorphic hardware prototyped by IBM achieved an extremely low energy consumption of $45pJ$ per spike for data communication [3].

However, conventional CMOS implementation of synapse design suffers from large area, high power consumption, and low-level parallelism. In *TrueNorth*, one digital neuron circuit is multiplexed to model 256 neurons, and a 256Kb SRAM is used for neuron data storage. Another SRAM for synapse weight works in parallel to update the neuron information, each synapse weight can only be chosen from one of -2, -1, 0, 1 and 2. As a result, the speed, size and bit-accuracy of neural network are limited by the SRAM: not only the sequential update of SRAM for neuron data storage limits the final system speed, but also the SRAM for synapse weight has very limited bit-accuracy and both together take a major part of the chip area. Last but not least, the majority of system energy was also consumed on SRAM reading/writing in recall and learning operations. [2, 4].

Memristor technology has gained significant attention in the neuromorphic system community since the first physical device demonstration in 2008 [5]. Electrical measurements demonstrated that memristors can produce many important features of biological synapses, such as *long term potentiation* (LTP), *long term depression* (LTD), *spike-timing-dependent plasticity* (STDP), and *short term potentiation* (STP) [6–15]. More importantly, the two-terminal structure of memristors enables the cross-point array implementation with high storage density and sub- pJ access energy [16]. Fully parallel recall operations through a cross-point array which applies synaptic weights to input signals and sends their summation to output neurons have been demonstrated in the latest research [17].

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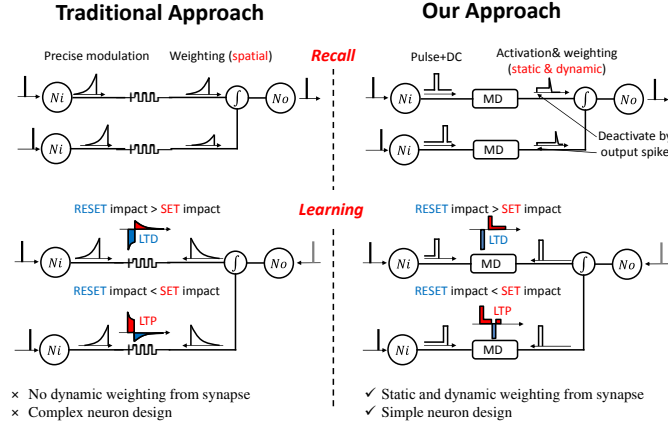


Fig. 1. A comparison between traditional approach and our new approach. Here, N_i represents input neuron, and N_o is used for output neuron.

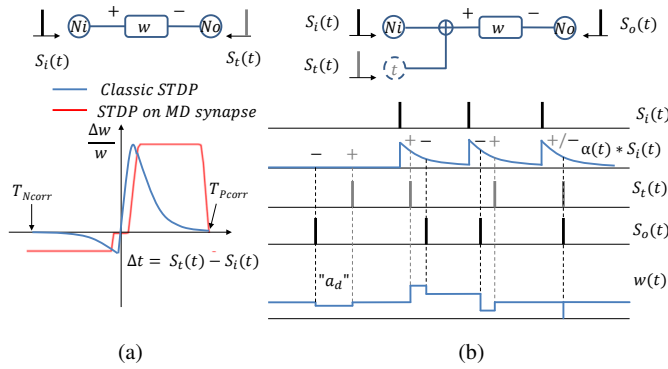


Fig. 2. Spike-timing-based learning rules and diagrams of neuron connections. (a) STDP learning rule; (b) ReSuMe learning rule.

Last but not least, comparing to state of the art digital implementations, memristor-crossbar-based system shows orders of magnitude better energy efficiency and speed in both inferencing [18] and training neural network algorithms [19]. Various neuromorphic circuits using single memristors to construct synapses have been presented [6–9]. However, these approaches are still far from mimicking synaptic models in spiking neural networks. For example, the popular α -function synaptic model [20] transforms an input voltage spike from pre-neuron to an α -function shaped current signal and passes it to post-neuron. The transformed current signal is a function of the input spiking time and the spatial weight of the synapse. Biological synapses exhibit even richer dynamic behaviors corresponding to more complex models [21,22]. In other words, a synapse cannot be simply taken as a static resistor but reveals dynamic property.

A common way of realizing both static and dynamic properties at software level is to integrate a dynamic synapse model into a spike-timing-based learning algorithm [1, 20, 23]. Currently, most of memristor-based synapse designs are based on multi-level/analog memristance value. Once learning is completed, the synaptic weight remains constant while the corresponding transformation function is performed in neuron circuit, through pulse shape modulator. As illustrated in Fig. 1, instead of spikes, a neuron circuit generates transformed sig-

nals to synapses [6–9]. Although many single-memristor based synapse designs demonstrated STDP feature theoretically and experimentally, the approach is not feasible for large-scale neural network implementation for the following reasons: (1) the neuron design complexity increases dramatically with the number of synapses, which is well known as *fanout issue*; and (2) it is difficult to preserve the quality of transformed signals that degrade rapidly as synapse number increases.

To enable the use of memristor-based synapse in large scale neural network implementation, we propose a novel design named as *memristor-based dynamic* (MD) synapse that provides the following features:

- *In-situ dynamic transformation function* determined by both the static weight and the relative dynamic spiking timing at pre- and post-neurons.
- *Two-terminal structure* that can be easily integrated in a cross-point array for massive connections.
- *Relaxation in spike signal requirement* which helps reduce neuron design complexity and improve communication energy efficiency.
- *Support of typical synaptic properties*, including spike-timing-based recall and synaptic weight tuning.
- *Spiking-timing based learning* demonstrated by utilizing fundamental STDP [23] and ReSuMe [20] learning rules.

Furthermore, we studied the use of MD synapses in SNNs by taking a temporal pattern learning task as example. Our results showed that the learning task can be accomplished at the nano-second scale with average $36.7pJ$ and $64.0pJ$ per learning spike for STDP and ReSuMe learning rules, respectively. The recall energy consumption is only $14.6pJ$ per recall spike without including the energy consumed by pre- and post-neurons. All the simulations were performed with the experimentally calibrated TaO_x memristor model [24], providing a strong device foundation for the results.

The paper is organized as follows. Section II gives a brief introduction on synapse and spiking neural networks; Section III describes the proposed MST synapse design and analyzes its electrical characteristics; Section IV validates the synaptic properties of the MST synapse design; Section V demonstrates its applications in a spiking neural network for a learning task and studies the design efficacy; Extended discussions regarding about the circuit implementation and the selection of memristor devices are presented in Section VI and Section VII, respectively; Section VII concludes the paper.

II. PRELIMINARY

A. Dynamic synapse

In human brains, synapses represent electrical and/or chemical communication links between neurons. It has been widely believed that the behavior of a biological synapse is a *spatio-temporal* process: a synapse is activated by input spikes from pre-neuron and transforms these spikes into decayed traces to post-neuron [25]. For example, the synaptic current from pre-neuron i to post-neuron j can be formulated through a popular α -function [26], such as

$$I_{i,j}(t) = w_{i,j} \cdot \int_0^\infty \alpha(s) S_i(t-s) ds, \quad (1)$$

where $w_{i,j}$ is the static synaptic weight describing the strength of the connection from pre-neuron i to post-neuron j . $S_i(t)$ is the input spike-timing train from pre-neuron i which can be represented as:

$$S_i(t) = \begin{cases} 1, & \text{if } t = t_{i,f} \\ 0, & \text{otherwise} \end{cases}, \quad (2)$$

where $t_{i,f}$ is the set of spiking times from pre-neuron i . The α -function can be defined as:

$$\alpha(t) = Ae^{-t/\tau_s}, \quad (3)$$

where A is the maximum magnitude of the decayed trace. And parameter τ_s is the synaptic time constant, which defines the duration of the decayed trace.

As can be seen, a dynamic synapse model contains two major factors: the (static) weight representing the synaptic strength and the dynamic function describing how a synapse is activated/deactivated by input spikes.

So far, the study of synaptic strength tuning mainly focuses on static weight w . The tuning of synaptic strength has two important attributes: *long term potentiation* (LTP) that represents a slow time constant strength potentiation after a synapse has received strong and positive stimulus from active connections, and *long term depression* (LTD) as an opposite process. The influence of LTP and LTD are strongly related to the spiking time, initial synaptic strength, and type of post-synaptic cells [27]. Another important property of biological synapses is their self-learning abilities, *e.g.*, *spike-timing-dependent plasticity* (STDP) [27]. Experiment on cultured hippocampal neurons showed that within a critical correlation time window, the post-synaptic spiking that peaked after synaptic activation caused LTP, whereas the post-synaptic spiking before synaptic activation, can result in LTD [27].

The synaptic weight modification is an important component in the learning process. *The Hebbian rule* and *The Delta rule* are two popular learning algorithms in *artificial neural network* (ANN) family. In The Hebbian rule, the synaptic weight between two neurons increase (decreases) when they activate simultaneously (separately). The Delta rule, also called as *least mean square* (LMS) method or Widrow-Hoff rule [28], further improves the training performance by utilizing the output signal of post-neuron as a feedback to minimize the error between the target and output signals. As the synaptic weight modifications are determined solely by current spike combinations, The Hebbian and The Delta rules are considered as static learning algorithms.

In spiking neural networks, *supervised STDP rule* can be regarded as an improved version of The Hebbian rule in temporal space as it considers the time correlation between the input and target spikes in modifying the synaptic weight. *ReSuMe algorithm* improves supervised STDP rule's performance which can further minimize the temporal difference between the target and output spikes. Both STDP and ReSuMe are temporal learning algorithms and more details of its use in our case study shall be described in Section II.C.

B. Synapse Design in Neuromorphic Circuits

To develop brain-like computing system, people need to build synapse hardware instead of simulating them. Neuromorphic circuits tend to mimic biological models and employ the same operation flows in weighting, tuning, and learning processes [29]. Previously, SRAM, floating gates, capacitors have been used in developing synapses [30]. These designs are severely constrained by the parallelism scale and the implementation size [3, 4]. Another prominent trend is designing synapses and neurons in analog or mixed-signal format [31]. The process variations and signal fluctuation potentially have great impacts on the system performance, which however can be partially amortized by synapse's learning ability [32].

The recently re-discovered *memristor devices* at nanometer scale [7] demonstrate synapse-alike behaviors, offering a more efficient way to implement synapses. For instance, Snider *et al.* successfully realized LTP, LTD, and STDP in a TiO_2 device controlled by *pulse width modulators* (PWM) [6]. The similar function was obtained in Ag/Si memristor synapse too through *time division multiplexing* (TDM) [8]. Another example is 1T1R-based HfO_2 synapse with pulse shape filters in pre- and post-neurons [9]. However, these synapse designs only contain the static weighting function. The temporal transformation has to be integrated into post-neuron. In a large network where a signal is directed to multiple synapses, however, it is difficult to well preserve shaped pulses through synapses, as shown in Fig. 1. So all these demonstrations require complicated neuron circuits and are at small scale. A compact dynamic synapse design that can be integrated into cross-point array is essential for large-scale SNNs.

C. Spiking Neural Network with Temporal Codes

In the brain, external events are represented as spatio-temporal patterns of neural activity [33]. How information is encoded, however, is still not very clear. Usually rate coding and temporal coding are adopted in system development. The rate coding paradigm assumes that information is represented by the frequency of spikes while the encoding scheme of temporal coding uses the exact timing of individual spikes [34]. Substantial empirical evidence in neurology supports the existence of temporal coding [35–38]. While there is evidence for both rate coding and spike-timing based coding in the biological nervous system, some experiments suggest that in the human retina, visual signals from 10^8 photoreceptor cells are projected to 10^6 *retinal ganglion cells* (RGCs) in a form of spike trains (temporal codes) through *temporal encoding* [39]. Then these spikes can be processed and learned through *spike-timing-based recall* and *spike-timing-based learning* processes [40]. In this work, we take a temporal pattern learning task as a case study to examine the performance of MD synapse design. The following two main functions are required in the application:

Spike-timing-based recall: In spike-timing-based recall, the synapse model changes the temporal weight of a synapse (*e.g.*, activated or deactivated) through temporal transformation function, and enables the neural network to produce the desired

temporal patterns. The static weights of synapses remain unchanged during the entire recall process.

In hardware implementation, two approaches usually are adopted to realize the spike-timing function. One attempt is to integrate the temporal transformation function into neuron so synapse only has a constant static weight. Complex neuron circuitry, such as PWM and TDM components, are required to provide precise timing controls of temporal transformation function and ensure the quality of signal even it is connected to multiple synapses [6, 8]. The other approach of realizing both transformation function and static weighting in synapse can greatly simplify the neuron scheme and alleviate the signal propagation issue. The unclear static weight and complex synapse design are main concerns of the approach.

Spike-timing-based learning: In this work, we used two spike-timing-based learning algorithms—supervised STDP¹ and ReSuMe. In both algorithms, weight updating of a synapse is determined solely by its own weight and excitation signals at the current time step. Such a local spatio-temporal learning scheme can be easily added to hardware synapse network. Here, a simple structure of one synapse connecting two neurons is used as example.

In STDP learning process, the post-neuron is forced to fire the target spike train $S_t(t)$. If an input spike $S_i(t)$ from the pre-neuron occurs within a predefined *positive correlation time window* (T_{Pcorr}) before $S_t(t)$, such as $0 \leq \Delta t = S_t(t) - S_i(t) < T_{Pcorr}$, implying the two neurons have a positive correlation, the synaptic weight increases. In contrast, a connection of two negative correlated neurons within a *negative correlation time window* (T_{Ncorr}) shall be weakened. Among a variety of STDP curves [41], Fig. 2(a) is a typical dependence of the synaptic weight change rate on Δt . The learning efficacy of STDP greatly relies on the correlation of the input and target spikes. When a target spike is uncorrelated to any of the input spikes, the STDP learning process fails. Thus, many input neurons firing at various patterns help increase the learning capacity of neural networks.

Unlike STDP, the ReSuMe learning rule requires output spikes to participate in the learning process and introduces a cost function to minimize the error between the target and output spikes. As illustrated in Fig. 2(b), input spikes are first converted to decayed traces through transformation function and then combined with the target spikes. The output spikes from post-synaptic neuron will be backpropagated to assist the weight updating. As such, the change of synaptic weight w from pre-neuron N_i to post-neuron N_o can be described as follows:

$$\frac{d}{dt}w(t) = [S_t(t) - S_o(t)][a_t + \int_0^\infty \alpha(s)S_i(t-s)ds], \quad (4)$$

where a_t is a constant that helps speed up the learning process, and $\alpha(t)$ is the pre-mentioned α -function that converts spike signal $S_i(t)$ to decayed traces. Eq. (4) implies that the synaptic weight w is updated when $S_t(t) \neq S_o(t)$. The direction and magnitude of the weight tuning are determined by the

sign of $S_t(t) - S_o(t)$ and the convolution term $\alpha(t) * S_i(t)$, respectively.

III. MEMRISTOR-BASED SPATIO-TEMPORAL SYNAPSE

In this work, we propose a dynamic synapse design by leveraging memristor technology. It supports both static weighting and dynamic temporal transformation function. So we name it as *memristor-based dynamic* (MD) synapse. Note that the discussion in the work is limited to positive weights because a memristor, as a physical device, can represent only a positive value. Our previous work proved that a negative value can be obtained by combining two or more memristor devices [32].

A. Design Concept

The objective of our work is to realize the following two types functions in synapse design:

- *Spatial weighting* that modulates a spike through synapse. The static weight is tunable and adjustable in learning process whereas remains unchanged in recalls. In our design, one memristor is used to store the weight.
- *Temporal transformation function* reflects if a synapse is activated (ON) or deactivated (OFF), which is determined by the spiking times of pre- and post-neurons in learning/recall process. This is realized by a second memristor in MD synapse, which turns on only when the pre- and post-synaptic spikes arriving at proper timings. If the timing is incorrect, it remains turned off, preventing transmission of signals to the post-neuron.

Fig. 3(a) depicts a MD synapse design which is composed of two memristors (M_1 & M_2) and one resistor (R). M_1 realizes the temporal transformation and the conductance of M_2 (G_2) represents the static weight. For ease of explanation, we assume M_1 and M_2 have an identical device structure. Therefore, they have the same ON and OFF resistance states (R_{ON} and R_{OFF}) and the identical switching dynamics. Note that the proposed MD synapse design can be generalized by applying different memristors with different characteristics to M_1 and M_2 , as we shall discuss in Section VII-A.

To protect the static weight G_2 in recall, we connect M_2 with resistor R in parallel so that the majority of the voltage across the synapse applies to M_1 . Accordingly, the following inequality corresponding to the worst-case situation when M_1 and M_2 are respectively at ON and OFF states shall be satisfied:

$$R_{ON} > R_{OFF} \cdot R / (R_{OFF} + R). \quad (5)$$

Assume $R_{OFF} = k \cdot R_{ON}$ and $R = x \cdot R_{ON}$, Eq. (5) turns to $k + x > kx$, which indicates that $x \leq 1$ shall be satisfied because $k > 1$. Here, we set $x = 1$ and make $R = R_{ON}$. By adjusting the voltage across the synapse, M_1 can be switched alone without impacting M_2 . The state change of M_2 , however, is always associated with M_1 's switching.

Fig. 3(a) also illustrates the cross-section view of a MD synapse structure: M_2 and R are isolated by a thin insulator and stacked above M_1 . The structure is very similar to a double layer *resistive random access memory* (ReRAM) device. The layout in Fig. 3(b) shows that MD synapse has an area of

¹In the rest of the paper, we refer STDP to supervised STDP learning rule for simplicity.

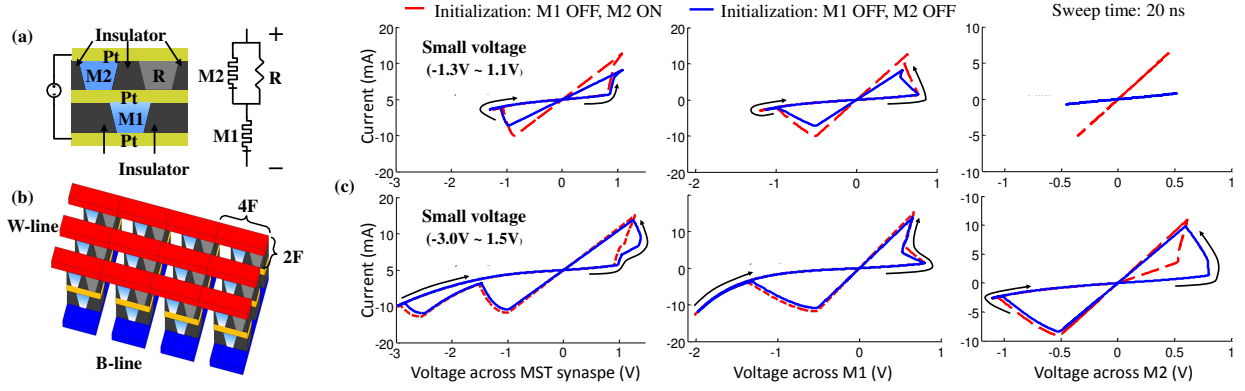


Fig. 3. Design and I-V characteristics of MD synapse. (a) Structure and equivalent circuit; (b) Conceptual layout in cross-point array; (c) DC-sweep I-V curves of MD synapse, M_1 , and M_2 when applying a small voltage (top figures) or a large voltage (bottom figures).

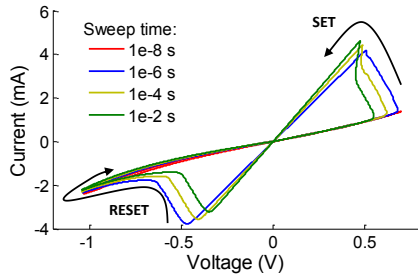


Fig. 4. The I-V curve of a TaO_x device. Reproduced based on [24].

$8F^2$, where F represents the technology feature size. Very importantly, the two-terminal structure can be easily integrated into cross-point arrays that have been widely investigated and explored for high-density synapse networks.

B. Characterization of MD Synapse

The proposed MD synapse generally is suitable for many types of memristor technologies. In the work, we adopted an experimentally calibrated model based on TaO_x devices [24]. TaO_x has been identified by HP Labs as one of the most promising memristor material for high density, low power, high endurance, and fast sub-ns programming [7, 42, 43]. Though TaO_x devices in nanometer dimension have been demonstrated [44, 45], we use the micrometer device model [24] in the following simulations for it is more mature and accurate at the current stage. The design concept of our synapse, however, can be extended to nano-devices.

Fig. 4 shows the I-V characteristic of the TaO_x device model. A sufficient positive SET pulse (negative RESET pulse) makes the TaO_x device switch toward ON (OFF) state. The SET/RESET switching threshold changes with the sweeping time, which has been considered in our simulations.

1) *DC Response*: We analyze the I-V characteristic of MD synapse and show the simulation results in Fig. 3(c). The corresponding I-V curves of M_1 and M_2 during the operation are also included for reference. Obviously, when the voltage across the synapse is too small to meet TaO_x device's SET/RESET threshold under a given sweep time, neither M_1 nor M_2 can

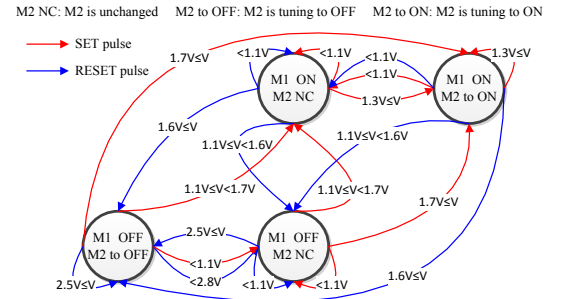


Fig. 5. The state transition diagram of MD synapse under spiking excitations. NC indicates that the device state is not change.

change. As the synapse voltage increases gradually, M_1 will first reach its switching condition. The top three I-V curves in Fig. 3(c) demonstrate that the dynamic component M_1 can be freely switched ON/OFF while the static component M_2 remains unchanged. The bottom three I-V curves in Fig. 3(c) implies that an even larger voltage amplitude is required to trigger the two-stage switching behavior including both M_1 switching and M_2 weight tuning.

In brief, a positive excitation that only set M_1 will enable the synaptic weighing function, while a negative excitation that only reset M_1 will disable the weighing function by putting the synapse back to high resistance state. To tune the weight of a MD synapse, the amplitude of the applied voltage shall be sufficiently large to switch M_1 and tune M_2 .

2) *Response to Spiking Excitation*: For its high scalability and energy efficiency, spiking neural network is good for large-scale neuromorphic implementation [4]. Instead of the complex synaptic signals in modulated format illustrated in Fig. 2, simple square pulses can be used for MD synapses as the input spikes at pre-neuron and the output spikes at post-neuron. To be consistent with TaO_x operation, we use SET/RESET pulses to represent the positive/negative voltage across a MD synapse.

Fig. 5 summarizes the state transition diagram of MD synapse by varying SET/RESET pulses at different voltage amplitudes.

The pulse duration is fixed at 500ps. For example, when

M_1 is ON, a RESET pulse with an amplitude between 1.1V and 1.6V can reset M_1 to OFF without changing M_2 's state; if the RESET pulse is larger than 1.6V, not only M_1 will be reset to OFF, but also M_2 can be tuned toward OFF.

Considering that the dynamic component M_1 is either ON or OFF² while the static component M_2 could be unchanged (NC), tuning to ON, or tuning to OFF, a synapse has six possible transition states. Based on the analysis in Fig. 3(c), the situations of “ M_1 ON & M_2 is tuning to OFF” and “ M_1 OFF & M_2 is tuning to ON” can never occur and hence are excluded. Notably, M_2 's tuning rate is not a constant value but determined by the present dynamic state and static weight of synapse as well as the applied voltage.

IV. SYNAPTIC FUNCTIONS OF MD SYNAPSE

In this section, the synaptic properties of MD synapse including the spike-timing-based recall and synaptic weight tunability will be examined and verified. Furthermore, STDP as the fundamental spike-timing-based learning rule will be applied to study MD synapse's learning characteristics. MD synapse could operate with pulses of milliseconds width to mimic the biological synapse behavior, however to pursue the performance potential, we use sub-nanoseconds pulses to demonstrate its functionalities. By integrating both transformation function and static weighting into synapses, neuron's interface can be simplified as spikes rather than precisely shaped pulses, as shown in Fig. 1. This also reduce the circuit design complexity of the neurons. Moreover, spikes signals are easier to propagate among a large scale of neural networks comparing to shaped pulses, thus MD synapse can also alleviate the signal propagation issue.

A. Spike-timing-based Recall

In a spike-timing-based recall, M_1 switches ON or OFF based upon the timing of pre- and post-spikes, performing a simple dynamic temporal transformation function. Accordingly, the synapse is activated or deactivated. Pulse shape modulator is not required because the pulse shaping effect is achieved through SETting/RESETing M_1 . For instance, Fig. 6(a) shows that a pre-spike from the pre-neuron initializes a positive SET pulse through the synapse, making it activated. A post-spike from the post-neuron results in a negative RESET pulse, which eventually deactivates the synapse. The static component M_2 remains at its initial value during the entire procedure.

A low DC signal of 0.2V is as the background signal in recall to support the transformation function. And the static weighting is reflected by the overall volume of charge through the synapse. Once a pre-spike arrives and enables M_1 , we start to track the accumulated charge. The assessment continues until a post-spike disables M_1 .

There are three types of typical timing situations in spike-timing-based recall. We examined and studied the MD synapse behavior for each case separately.

²For simplicity, we refer M_1 's ON/OFF state as the MD synapse state in the rest of this paper.

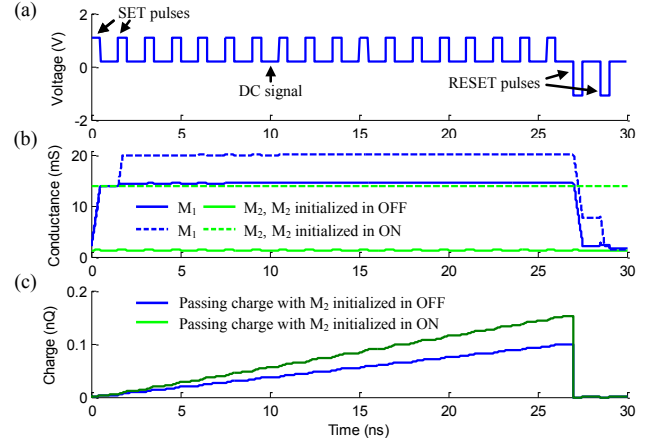


Fig. 6. MD synapse in spike-timing-based recall – Case 1.

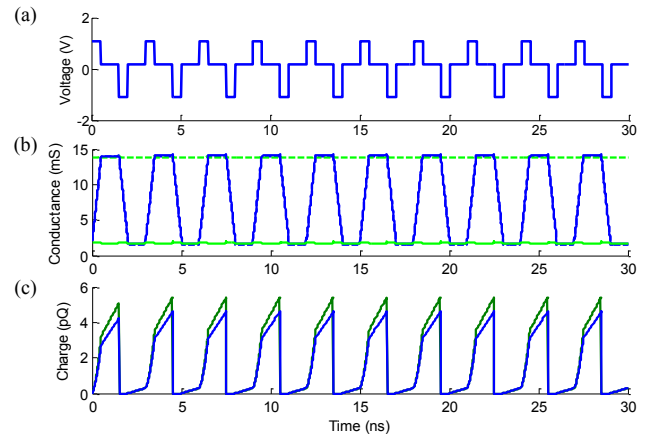


Fig. 7. MD synapse in spike-timing-based recall – Case 2.

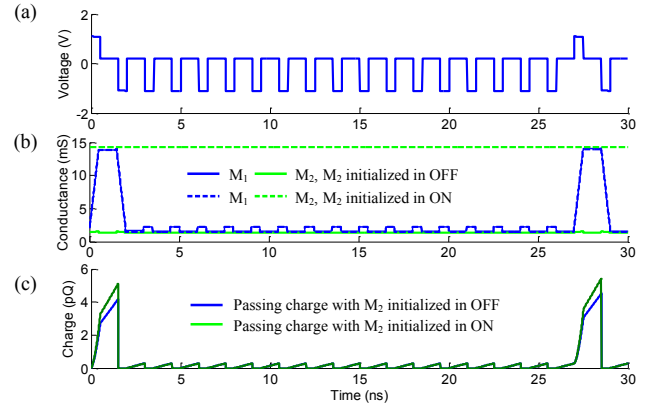


Fig. 8. MD synapse in spike-timing-based recall – Case 3.

Case 1: The pre-neuron fires much faster than the post-neuron, forming a multi-spike train with many SET pulses followed by a few RESET spikes. The MD synapse behavior under such a condition is shown in Fig. 6. M_1 turns ON at the first SET pulse and remains ON until a RESET pulse comes. Because of the over-tune induced memristance shift [46], the conductance of M_1 could increase further if two SET pulses fire in consequence. Even though, M_1 can still be switched OFF by extending the duration or number of RESET pulses. Notably, during the entire recall process, the resistance of M_2

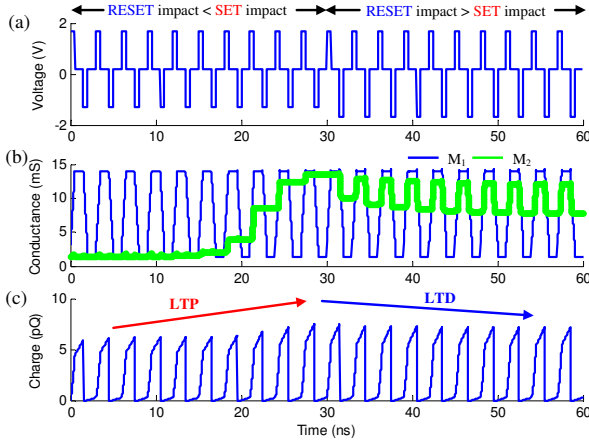


Fig. 9. The tunability of MD synapse. In the first 10 cycles, $V_{SET} = 1.7V$ and $V_{RESET} = -1.3V$; In the following 10 cycles, $V_{SET} = 1.7V$ and $V_{RESET} = -1.7V$. The pulse duration is set to 500ps.

remains constant and determines the total charge through the synapse. More specific, the charge accumulation is faster when M_2 is R_{ON} but much slower if M_2 is R_{OFF} .

Case 2: When the pre- and post-neurons are strongly correlated, their firing events occur alternatively and appear as a sequence of SET/RESET pulses through the synapse. Under the circumstance, M_1 (and the synapse) switches between ON and OFF states but M_2 is not affected, as shown in Fig. 7. The charge accumulation through the synapse is at a low rate when the synapse is OFF but increases significantly once the synapse turns on. This process is robust and repeatable.

Case 3: A backward multi-spike train is generated when the burst of the post-neuron is much faster and more frequent than that of the pre-neuron. As the opposite situation of Case 1, the spike-train through the synapse is in a form of many RESET pulses followed by several SET pulses. During the sequence of RESET pulses, the synapse is OFF and thus not much charge can pass through it. Certainly, M_1 can be re-activated by any SET pulse and M_2 remains intact.

B. Weight Tunability

As the basis of learning process, the weight tunability of MD synapse including LTP and LTD was examined. Note that the tuning targets at only the static weight of M_2 , not for M_1 that represent the activation status of the synapse. As shown in Fig. 5, spikes with large amplitude at the pre-/post-neurons (or strong SET/RESET pulses) enable the tuning process.

For ease of explanation, we give an example shown in Fig. 9. Assume a strong correlation exists between pre- and post-neurons. They fire alternatively, appearing as a sequence of SET/RESET pulses through the synapse. In the first 10 cycles, the spikes generated at pre-neuron is stronger. As a result, M_2 gradually shifts toward ON state with better conductivity, successfully demonstrating the LTP feature. The scenario in the following 10 cycles is opposite: the stronger spikes at the post-neuron makes the RESET procedure more efficient. Thus, the effective conductance of M_2 when the synapse turns ON gradually reduces, implying a LTD behavior. The change of

MD synaptic strength (conductance) is reflected by the charge passed through the synapse, as shown in Fig. 9(c). In summary, the positive stimuli corresponding to *stronger-SET:weaker-RESET* combination enables LTP feature. And the LTD can be realized under the circumstance of *weaker-SET:stronger-RESET* pulses, implying negative stimuli. Note that M_1 takes majority of pulse voltages and hence always reach ON or OFF state. In contrast, M_2 's conductance slowly changes because only a small amount of voltage applies on it.

As seen in Fig. 9(b), the conductance of the synapse increases in “steps” and decreases in “hops”. This is because a constant SET voltage $V_{SET} = 1.7V$ is applied in all the cycles. The conductance of synapse is always increases by a SET pulse and then decreased more by a RESET pulse. We particularly selected and presented this fast timing setting to investigate the performance limitation of TaO_x devices. Moreover, an asymmetric tuning curve of M_2 can be observed: It takes 10 pulses to sweep M_2 from OFF to ON, while the switching in the opposite direction is much slower due to intrinsic switching asymmetry in TaO_x devices [24].

C. STDP Learning

As an improved version of The Hebbian learning rule at temporal space, STDP learning rule can be taken as a causality detector. Correlation time window of a spike is used to evaluate its causality with other spikes: if a pre-spike fires before (after) a post-spike within the correlation time window, the synaptic strength of the synapse in between shall be potentiated (depressed), corresponding to a LTP (LTD) behavior. Here STDP learning rule is supervised so the post-spike is forced to fire at the target spike pattern.

Fig. 10 shows the realization of STDP learning by employing MD synapse under these two conditions. Here, a pre-spike is followed by a small DC signal of 0.4V lasting for T_{Pcorr} , representing the positive correlation time window of the spike. The negative correlation time window T_{Ncorr} can be formed by setting a normal RESET pulse $T_{Pcorr} - T_{Ncorr}$ ahead of the post-spike. In this way, synapses for uncorrelated input spikes are pre-deactivated and will not be affected. Since a natural T_{Ncorr} has been naturally defined by the time between previous post-spike and the current pre-spike in the design, a separate setting of T_{Ncorr} can be saved.

Fig. 10(a) illustrates the scenario when the pre-spike injects first and the corresponding post-spike falls within T_{Pcorr} of the pre-spike. Though the *strong* SET pulse(pre-spike) remain unchanged, the small DC signal associated with the pre-spike degrades the *strong* RESET pulse(post-spike) to a *normal* RESET. Such a condition makes the M_2 conductance increase, resulting in a LTP process. The simulation results match well with our previous analysis in Section III-B. The realization of LTD process is shown in Fig. 10(b), where the RESET pulse (post-spike) remain *strong* and has larger impact.

Causality decision is not the only important criteria of a synapse's STDP learning ability, which is also determined by the amplitude, time and shape of spikes as well as the initial synaptic weight. We characterize the learning efficiency of a MST synapse. The characterization is conducted by applying

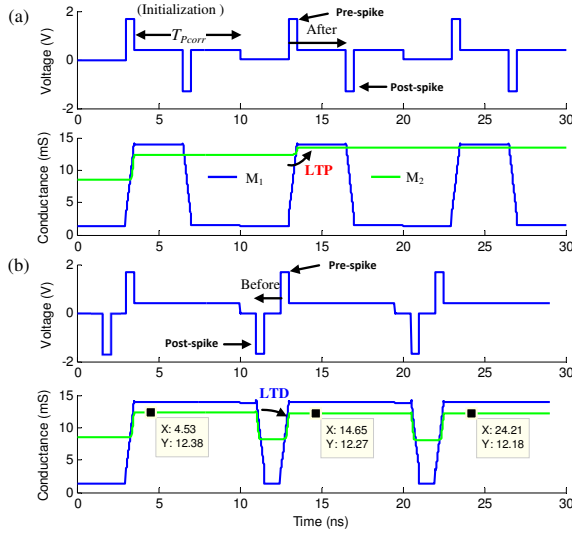


Fig. 10. Employ MD synapse to realize STDP learning. (a) LTP; (b) LTD.

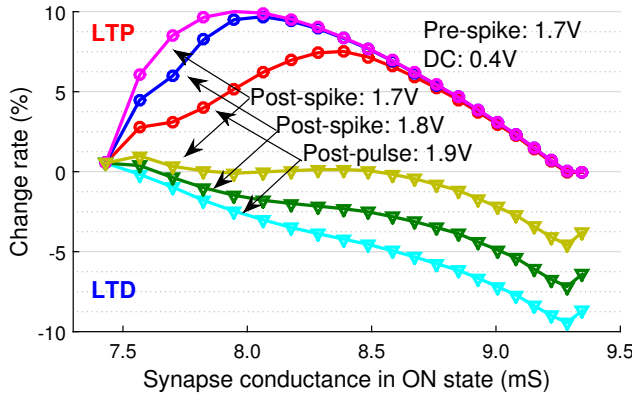


Fig. 11. The conductance change rate vs. the initial synapse state for STDP.

two recall cycles respectively before and after a STDP learning cycle (either LTP or LTD) at different pre-/post-spike configurations [27]. The learning efficiency is measured by the change of effective synapse conductance (activated) in the two recalls.

The results in Fig. 11 show that for LTD, the change rate of synapse conductance increases as the initial state of synapse is more conductive. This reflects the M_2 's switching feature from ON to OFF, as shown in Fig. 9(b). The LTP, however, demonstrates an interesting convex curve. It implies that during M_2 's switching process from OFF to ON, its conductance changes faster at the beginning and then slows down as the device finally turns saturated at a full ON state. The difference in LTD and LTP characterization is resulted by the asymmetric switching of TaO_x memristor.

D. ReSuMe Learning

ReSuMe learning rule is the temporal version of Delta learning rule. It has better learning quality than STDP since it uses the free post-spikes as feedback to minimize the error between target spikes and post-spikes.

We use the example in Fig. 12 to illustrate the ReSuMe learning ability of MD synapse. Similar as STDP learning, a

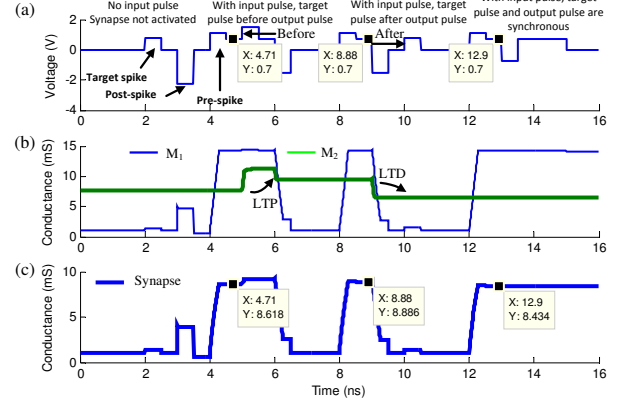


Fig. 12. Employ MD synapse to realize ReSuMe learning.

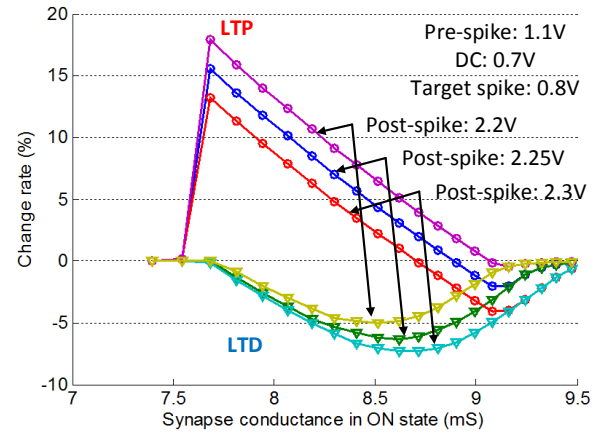


Fig. 13. The conductance change rate vs. the initial synapse state for ReSuMe.

small DC signal with a period of T_{corr} following a pre-spike is used to represent its correlation time window. However, based on the requirement of ReSuMe, the DC signal shall terminate right away when post-spike is generated. Thus, the convolution term $a_{di}(t) * S_i(t)$ of the original ReSuMe algorithm in Fig. 2 can be represented by the synaptic current excited by the pre-spikes and the following DC signals. The given example presents four typical situations.

- 1) There is no pre-spike. Since target spike is lower than SET threshold, MD synapse remains at OFF state. Neither the target spike nor the post-spike can change the synaptic conductance.
- 2) The target spike happens before the post-spike, and both of them fall into the correlation window of the pre-pulse. As such, the post-spike performs as a normal RESET and causes a LTP process.
- 3) Similar to scenario 2), however, the post-spike happens before the target spike. The post-spike tunes M_2 toward OFF state, implying a LTD process. Under this situation, the target spike does not contribute to the learning process because it cannot SET the device alone while the DC signal has already been terminated by post-spike.
- 4) The target spike and post-spike are approximately synchronized. There is no update on M_2 , the synapse keeps

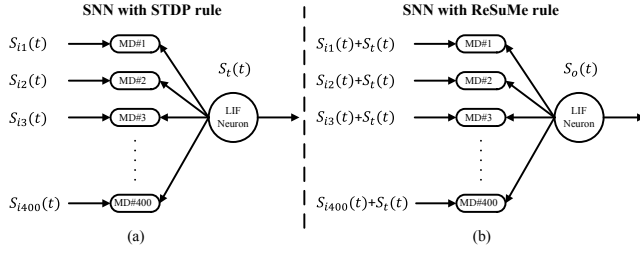


Fig. 14. A spiking neural network with 400 synaptic inputs and 1 LIF output neuron and the spike configurations for (a) STDP rule or (b) ReSuMe rule.

at ON and the DC signal applies. Note that the design even do not require a perfect matching of the target spike and post-spike because memristor state change requires SET/RESET pulse last for sufficient time.

There are two situations missing in Fig. 12. When the target spike is uncorrelated to the pre-spike, it won't affect the synaptic weight at all. Or, if the post-spike does not meet the time correlation window, it behaves as a strong RESET pulse resulting in a LTD process. Fig. 13 summarizes the ReSuMe learning characteristics of MD synapse. The result was obtained through the similar characterization flow in Section IV-C while the recall cycles were applied between the learning cycles. Particularly, we examined the synaptic conductance in ON state. With larger post-spike as RESET, the LTP and LTD curves shift down linearly. Therefore, ReSuMe exhibits the better learning characteristics than STDP – better linearity of peak current change rate and more flexible state modification.

Interestingly, both STDP's and ReSuMe's learning characteristics obtain the similar trend as the observation in biological synapses: LTP shows a linear decreasing trend with synaptic strength's increment, while LTD is not strongly relevant with initial synaptic strength [27]. For TaO_x-based MD synapse, ReSuMe's LTP behavior has a better linearity than STDP's, while the characterization of LTD demonstrates an opposite trend. This can be explained by the intrinsic asymmetric ON and OFF switchings of TaO_x devices [24].

V. MD SYNAPSES IN SPIKE-TIMING NEURAL NETWORK

A. Application Setup

We demonstrate the use of MD synapses in spiking neural network by adopting a case study from [20]. The selected neural network is trained on a random target firing pattern of 400ns. As shown in Fig. 14, 400 synaptic inputs go through 400 MD synapses and then are collected by a *leaky integrate-and-fire* (LIF) neuron. The temporal encoding method (Section II-C) was applied to convert any static pattern into spatio-temporal patterns as the input spiking trains $S_{i,1\sim 400}(t)$ in the given study case. If MD synapses have been properly trained, the fire pattern of the output neuron $S_o(t)$ shall converge to the target firing pattern $S_t(t)$.

To accommodate the easy timing control of MD synapses, the function of the LIF neuron can be simplified as follows: (1) the charge at the LIF neuron is leaking at a constant speed of

TABLE I
SIGNAL SETUP FOR SPIKING NEURAL NETWORK

	V_{SET}	V_{RESET}	V_{TARGET}	V_{DC}	T_{pulse}	$T_{P_{corr}}$
Recall	1.1V	1.3V	–	0.2V	0.5ns	9.5ns
STDP	1.7V	1.8V	–	0.4V	0.5ns	9.5ns
ReSuMe	1.1V	2.2V	0.8V	0.7V	0.5ns	9.5ns

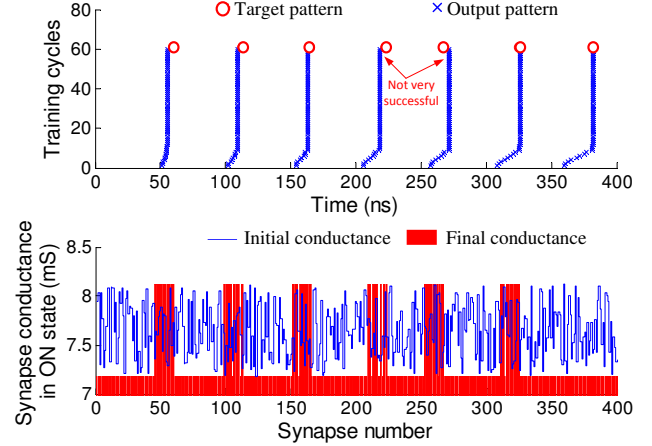


Fig. 15. The effectiveness of temporal pattern learning by using STDP.

Q_{leak} ; and (2) the LIF neuron fires whenever its accumulated charge exceeds the firing threshold, *i.e.*, $Q > Q_{th}$.

For ease of comparison, we adopt the same assumption in [20] that each synaptic input fires only once during the single presentation of the target signal. And the particular input pulses are distributed uniformly throughout the 400ns time interval. The conductances of 400 MD synapses are initialized randomly by applying a Gaussian distribution to M_2 conductance. All the synaptic inputs are limited to excitatory because the MD synapses based on real physical devices does not provide negative weights. Table I summarize the signal setup in recall and learning processes.

B. Learning Temporal Patterns Using STDP

The STDP learning rule was applied to the given application. As illustrated in Fig. 14(a), during the learning process, the LIF neuron is forced to fire at the target pattern $S_t(t)$. Fig. 15(a) shows the progress of 60 learning cycles, each of which last 400ns. Within each learning cycle, the target pattern $S_t(t)$ fires at the designed time illustrated by $\circ$ in the figure. After each learning cycle, we conduct a recall and record the output pattern $S_o(t)$ as $\times$. The simulation results show that overall $S_o(t)$ tends to approach $S_t(t)$ but it is not always successful. This is due to the intrinsic drawback of STDP learning rule: the output of neuron does not participate in the learning process. So sometimes the difference between $S_t(t)$ and $S_o(t)$ cannot be minimized.

Fig. 15(b) compares the conductance of the 400 MD synapses at the initialization (blue curve) and after 60 cycles of learning (red bars). At the end of learning process, most synapse are saturated at conductance boundaries. This is mainly because STDP learning rule requires relatively large synaptic weight range [47]. The scenario also implies that

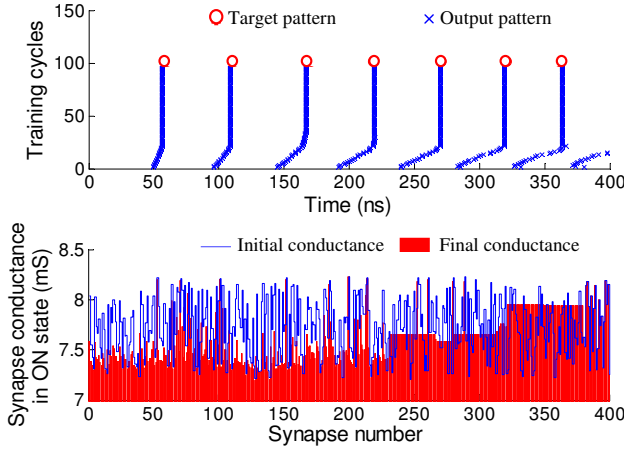


Fig. 16. The effectiveness of temporal pattern learning by using ReSuMe.

the constrains of physical memristor devices could greatly affect the learning performance. After all, a large range of memristance could improve the STDP learning performance.

C. Learning Temporal Patterns Using ReSuMe

Similarly, we examined the effectiveness of temporal pattern learning of the given design using ReSuMe learning rule [20]. Different from STDP, the ReSuMe learning rule includes the error minimization between $S_t(t)$ and $S_o(t)$. As shown in Fig. 14(b), $S_t(t)$ is combined with every synaptic input signal $S_i(t)$ to generate $S_o(t)$ at the LIF neuron. The output pattern $S_o(t)$ is then fed back to every synapse to participate in the learning process. Since the memristor state change relies on both the amplitude and period of the excitation, an error margin can be naturally formed between $S_t(t)$ and $S_o(t)$: whenever the target and the output pulses overlap, the rest of their pulse duration has little impact on the memristor state.

The learning progress of ReSuMe rule of the given example is shown in Fig. 16(a). In most test cases, the output pattern generated by the LIF neuron converges to the target pattern within 25 learning cycles, demonstrating a much successful learning ability. Moreover, the pattern learning tasks can be completed without observing synapse conductance saturation as shown in Fig. 16(b). It delivers an important information: even the MD synapse design based on TaO_x devices has limited synaptic conductance tuning range, it can still provide sufficient learning ability in neural network applications with assist of appropriate learning rules. Further increase of memristance range will alleviate the situation.

D. Energy Estimation

To evaluate the energy efficiency of MD synapses, we calculate the energy consumption per spike in recall and learning processes by following

$$E_{\text{total}} = E_{\text{SET}} + E_{\text{DC}} + E_{\text{RESET}}, \quad (6)$$

where, $E = V^2 T_S / R_{\text{MD-synapse}}$. T_S is the duration when the signal is applied. An additional energy consumption related to the target pulse E_{TARGET} shall be added into Eq. (6) in

ReSuMe learning. Here, only the energy consumed on MD synapse is estimated. Based on the signal setup in Table I, the recall consumes only 14.6pJ per spike and the energy consumption in learning process is 36.7pJ and 64.0pJ per learning spike for STDP and ReSuMe, respectively. Note that the micro model of TaO_x devices used in this work has a low resistance range of 70Ω ~ 670Ω [24]. Applying nano-scale devices can significantly increase the memristance value to 100KΩ ~ 1MΩ [7] and further reduce energy consumption. For instance, the micro-scale TaO_x device needs >10 mA current to switch, but for the nano-scale device, the switching current will be at only ~10 uA level, which indicates a 3 orders of magnitude of energy reduction and the MD synapses could be operated at several fJs. Comparing to other analog synapse components, a nano-scale MD synapse can provide more complex weighting function with equal or less energy consumptions as well as simplify the neuron design. [48].

VI. NON-IDEAL CONDITIONS IN IMPLEMENTATION

Effectively connecting neurons is an essential requirement in neural network hardware implementation. For a fully-connected network with n neurons, the number of synapses is in an order of n^2 . The memristor cross-point array illustrated in Fig. 3(b) is expected to provide the densest implementation and the best connectivity of synapse network. However, the non-ideal conditions in circuit implementation, such as the device variations and signal degradation, may severely affect the system performance. The impacts on the MD synapse based cross-point arrays will be discussed in this section.

A. The Impact of Device Variations

In the study, the variations of all the three devices within the MD synapse, including M_1 , M_2 , and R , are considered in the experiment. We assume that the variation of each device follows a Gaussian noise distribution with σ up to 1% to 10%. Fig. 17 presents the induced variations of V_{m1} and V_{m2} (the voltages across M_1 and M_2 , respectively) under two worst-case scenarios: (a) in RESET process, M_1 is in ON state and M_2 is in OFF state; and (b) in SET process, both M_1 and M_2 are OFF. As aforementioned, to ensure the proper function of MD synapse, M_1 shall always have a higher switching priority than M_2 . Accordingly, the device variation $\sigma \leq 6\%$ is tolerable in MD synapse design. As σ is greater than 6%, it is possible that V_{m1} is lower than V_{m2} in RESET process, violating the operation condition of MD synapse.

B. The Impact of Signal Degradation

In a typical cross-point array structure, the signal degradation is mainly affected by two factors: the most conductive unselected devices which sink a large portion of current flow (so called as *sneak current*), and the resistance of metal wires used in cross-point array. More specific, the ratio of the ON state resistance of the cross-point devices over the wire segment resistance R_w limits the array scale [49].

Fig. 18 shows the effective voltage of cross-point device normalized to the input pulse voltage under the worst-case

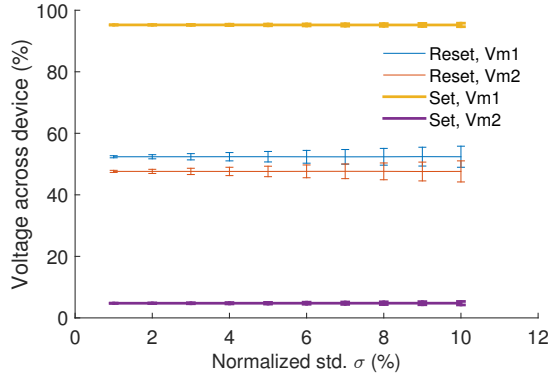


Fig. 17. The impact of device variations with normalized standard deviation σ under the worst SET and RESET scenarios.

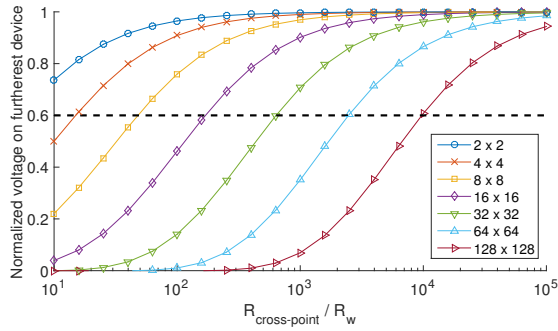


Fig. 18. The impact of wire resistance on effective voltage across the device under the worst-case scenario.

operating condition defined in [49]. For a conventional cross-point array with a TaO_x memristor as the cross-point device, $R_{ON}/R_w \approx 100$. To ensure that the furthest corner device can reach at least 60% of the input voltage signal, the array size cannot exceed 8×8 . When developing the cross-point array with the MD synapse design, the lowest resistance bond of unselected (deactivated) MD synapses is

$$\min R_{\text{cross-point}} = R_{M1, \text{OFF}} + \frac{R \cdot R_{M2, \text{ON}}}{R + R_{M2, \text{ON}}}. \quad (7)$$

Where $R_{M1, \text{OFF}}$ is the resistance of M_1 in OFF state, and the second term on the right side represents the parallel resistance of R and M_2 in ON state. For a TaO_x-based MD synapse with the given device parameters, $\min R_{\text{cross-point}}/R_w \approx 1000$. As seen in Fig. 18, the array size can be extended to 32×32 . Moreover, if different types of memristor technologies can be applied to M_1 and M_2 so as to increase $R_{M1, \text{OFF}}$ only, $\min R_{\text{cross-point}}$ can further increase. The scale and performance of MD synapse can dramatically improve.

VII. MEMRISTOR DEVICES IN MD DESIGN

A. MD Synapse with Two Different Memristors

In original MD synapse design, two identical memristors are employed so a resistor R is needed to protect M_2 in recall process. Utilizing two different types of memristor devices for M_1 and M_2 and eliminating the parallel resistor can result in a more efficient design from two aspects: (1) the MD

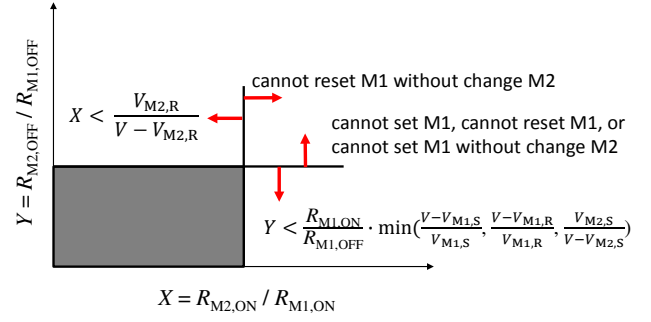


Fig. 19. Resistance ratio of M_1 and M_2 . The gray area is the working region that satisfies Eq. (8). X and Y specify the boundary conditions.

synapse structure is simplified to two memristors connected in series. The unit area of a synapse decreases to $4F^2$ so the fabrication cost reduces; (2) when M_2 is not forming free, the difficulty in M_2 's forming process lowers. If both memristors are voltage controlled, their device parameters, including the low resistance states (R_{ON}), high resistance states (R_{OFF}), SET voltage amplitudes (V_S), and RESET voltage amplitudes (V_R) shall satisfy the following constrains:

$$\begin{aligned} \frac{V_{M1, S}}{V_{recall}} &\leq \frac{R_{M1, ON}}{R_{M1, ON} + R_{M2, OFF}} && \text{when SET } M_1; \\ \frac{V_{M2, S}}{V_{recall}} &> \frac{R_{M1, ON} + R_{M2, OFF}}{R_{M1, ON}} && \text{when SET } M_1; \\ \frac{V_{M1, R}}{V_{recall}} &\leq \frac{R_{M1, ON}}{R_{M1, ON} + R_{M2, OFF}} && \text{when RESET } M_1; \\ \frac{V_{M2, R}}{V_{recall}} &> \frac{R_{M2, ON}}{R_{M1, ON} + R_{M2, ON}} && \text{when RESET } M_1. \end{aligned} \quad (8)$$

Here, V_{recall} is the pulse amplitude used in recall process.

The first two constrains correspond to the worst-case condition when setting M_1 . And the last two constrains describe the worst-case condition in resetting M_1 . In a recall process, M_2 shall always keep intact. Fig. 19 shows these constrains in a graphic view. The gray area represents the working region that satisfies these inequalities. Without loss of generality, we assume $V_{M1, R} > V_{M1, S}$. Thus, the minimal allowable V is

$$V = V_{M1, R} + V_{M2, S}. \quad (9)$$

and the optimal configuration of M_1 and M_2 shall obey the following relations to maximize the resistance ratio between M_2 and M_1 :

$$\frac{R_{M2, ON}}{R_{M1, ON}} \leq \frac{V_{M2, R}}{V_{M1, R} + V_{M2, S} - V_{M2, R}}, \quad \text{and} \quad (10a)$$

$$\frac{R_{M2, OFF}}{R_{M1, OFF}} \leq \frac{V_{M2, S}}{V_{M1, R}} \cdot \frac{R_{M1, OFF}}{R_{M1, ON}}. \quad (10b)$$

The two independent inequalities of Eq. (10) define the relation of M_2 's four parameters. For instance, with the assumption of $V_{M1, R} > V_{M1, S}$, the SET process of M_1 does not appear as the worst case to constrain the design, implying $V_{M1, S}$ doesn't have a strict requirement. In general, relatively low switching voltage and high resistance range are preferable for M_1 . The high SET voltage and low RESET voltage are more important for M_2 to improve the effective OFF/ON ratio and therefore get better synapse performance. Here, the effective OFF/ON ratio corresponds to the range of memristor resistance states to promise proper functionality of MD synapses.

In the case of current-controlled memristor devices, we can remove the resistor to let the same amount of current passes through M_1 and M_2 . In order to remain the state of M_2 intact when switching M_1 ON/OFF, M_2 shall be able to tolerate a larger switching current than that of M_1 .

B. The Use of Memristors with Volatile Switching

Memristor is usually regarded as a nonvolatile device. However, the retention loss has also been observed in WO_x and Ag_2 memristors under low and short pulse excitations [10, 11]. This indeed can be taken as a form of *volatile switching*, meaning, after a memristor switches its state, the conductance drifts back to the original state (or certain states) within a time scale from nanoseconds to minutes. In single-memristor based synapse designs, the volatile switching behavior has been linked to the *short term potentiation* (STP) [10, 11]. Opposite to LTP, STP describes the phenomenon in nervous system that under small excitations, synapse strength rises and then decays to the original value after the excitation is removed.

The proposed MD synapse contains two memristors. The volatile switching behavior of M_2 , if any, can also be related to STP as M_2 represents the spatial weight of synapse. If M_1 demonstrates the volatile switching behavior, the temporal transformation function of the synapse will be affected. As a simple illustration, let's assume M_1 's conductance exponentially decays over time after a volatile setting. It has two important design implications: First, solely utilizing the DC signal on synapses after SET pulse may be able to form a more complex and accurate transformation function, e.g., α -function in Eq. (3). Second, as M_1 can turn itself off, the RESET pulse from post-neuron will be no longer needed. The neuron design complexity can be further reduced.

VIII. CONCLUSION

In this paper, we present a *memristor-based dynamic* (MD) synapse design. By adopting the latest experimentally grounded TaO_x model, we characterize its electrical properties and synaptic features. The simulation results show that the design can enriches the synaptic functionalities in temporal recall and learning abilities. Meanwhile, the design complexity of neurons can be significantly reduced. Furthermore, we perform a temporal pattern learning task by utilizing STDP and ReSuMe learning rules, to demonstrate the usage of MD synapse in SNNs. Our result shows recall and learning ability in nano-second scale. The average recall energy is merely 14.6pJ per recall spike. And the learning energies when utilizing STDP and ReSuMe are 36.7pJ and 64.0pJ per learning spike, respectively. Comparably, the ReSuMe learning rule achieves better accuracy and faster learning speed than the STDP learning rule. The dependency on the constraints of memristor device physical parameters is also more relaxed when utilizing the ReSuMe learning rule. In addition, the simple structure of MD synapse naturally supports the integration of cross-point arrays and alleviates the complexity of the associated neuron design. These promising features make the hardware implementation of massive SNNs closer to reality.

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